PUCP

INGENIERÍA INFORMÁTICA

1INF52: Deep Learning



Tema 6: Transfer Learning

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Review

Keras code

```
from tensorflow.keras.layers import Dense, Conv2D, MaxPool2D, Flatten


model = Sequential([
    Conv2D(16, 3, activation='relu', input_shape=(28,28,1)),
        MaxPool2D(),
        Conv2D(32, 3,
        MaxPool2D(),
        Flatten(),
        Dense(10, act
])
```

¿Cuántos parámetros tiene esta capa de convolución?

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¿Cuántos parámetros tiene esta capa de convolución?

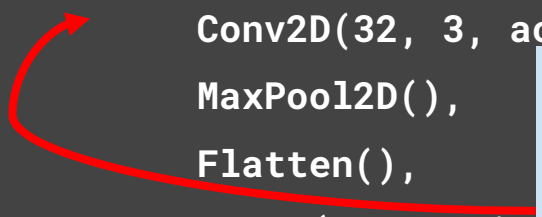
16 filtros de 3x3x1 -> 16x3x3x1 = 144

+ 16 bias -> 144 + 16 = **160**

Keras code

```
from tensorflow.keras.layers import Dense, Conv2D, MaxPool2D, Flatten

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        Flatten(),
        Dense(10, activ
])
```

¿Cuántos parámetros tiene esta capa de convolución?

32 filtros de 3x3x16 -> 32x3x3x16 = 4,608

+ 32 bias -> 4,608 + 32 = **4,640**

Keras code

```
from tensorflow.keras.layers import Dense, Conv2D, MaxPool2D, Flatten

model = Sequential([
    Conv2D(16, 3, activa
        MaxPool2D(),
        Conv2D(32, 3, ac
        MaxPool2D(),
        Flatten(),
        Dense(10, activation='softmax')
])
```

¿Cuántos parámetros tiene la capa Dense, si el output del último MaxPool2D tiene la forma Nx5x5x32?

Keras code

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from tensorflow.keras.layers import Dense, Conv2D, MaxPool2D, Flatten

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        MaxPool2D(),
        Flatten(),
        Dense(10, activation='softmax')
])
```

¿Cuántos parámetros tiene la capa Dense, si el output del último MaxPool2D tiene la forma Nx5x5x32?

Nx5x5x32 = Nx800

Nx800 @ 800x10 = Nx10

pesos: 800x10 -> 8,000

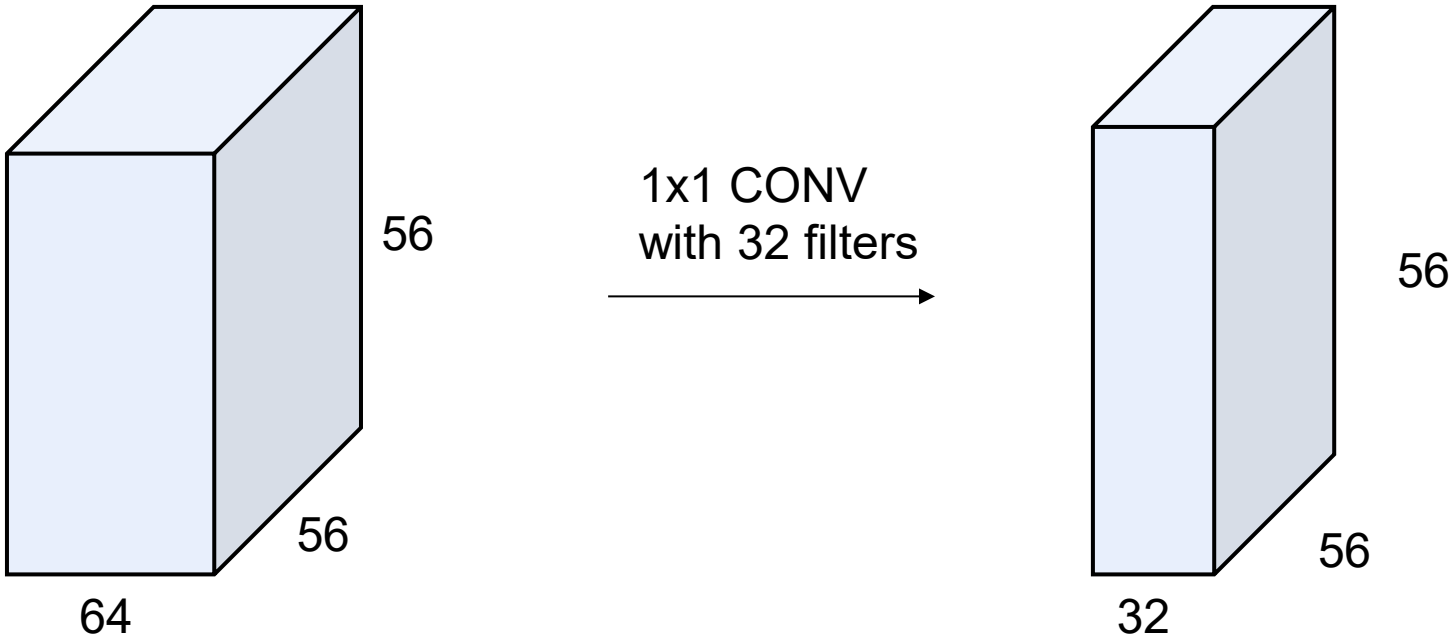
+ 10 bias -> **8,010**

Keras code

```
1  model.summary()
```

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 26, 26, 16)        160
_________________________________________________________________
max_pooling2d (MaxPooling2D) (None, 13, 13, 16)        0
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 11, 11, 32)        4640
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 5, 5, 32)          0
_________________________________________________________________
flatten (Flatten)            (None, 800)               0
_________________________________________________________________
dense (Dense)                (None, 10)                8010
=================================================================
Total params: 12,810
Trainable params: 12,810
Non-trainable params: 0
_________________________________________________________________
```

1x1 convolution layers



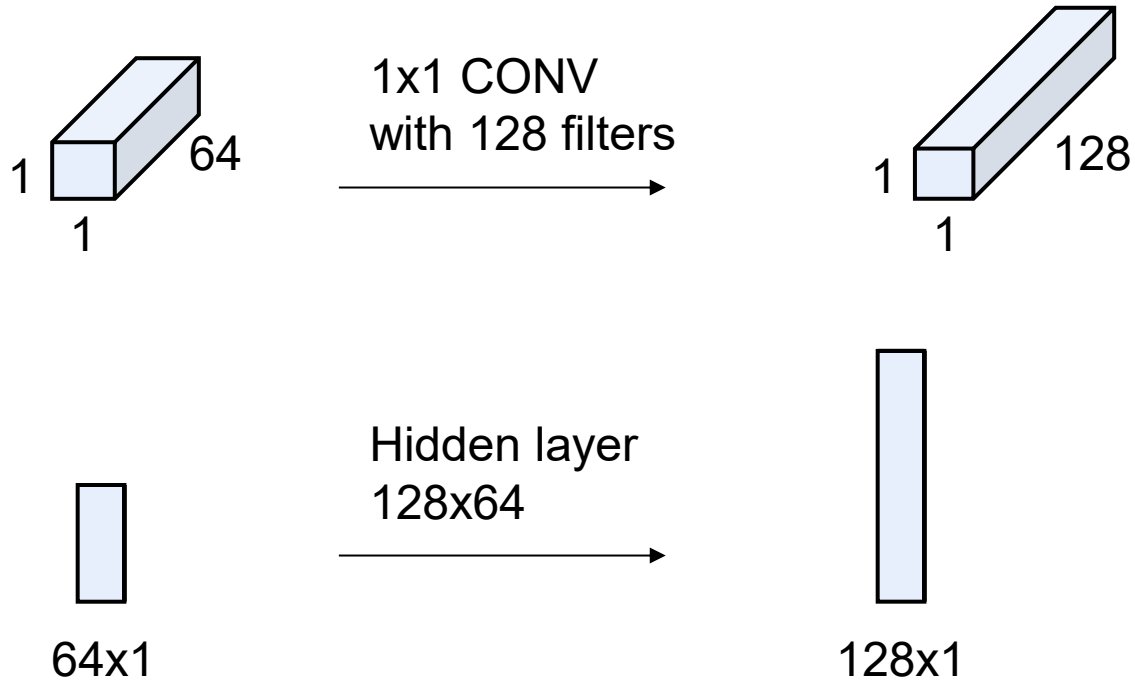
1x1 convolution layers

Ej: 128 filtros de 1x1 en un input de 1x1x64

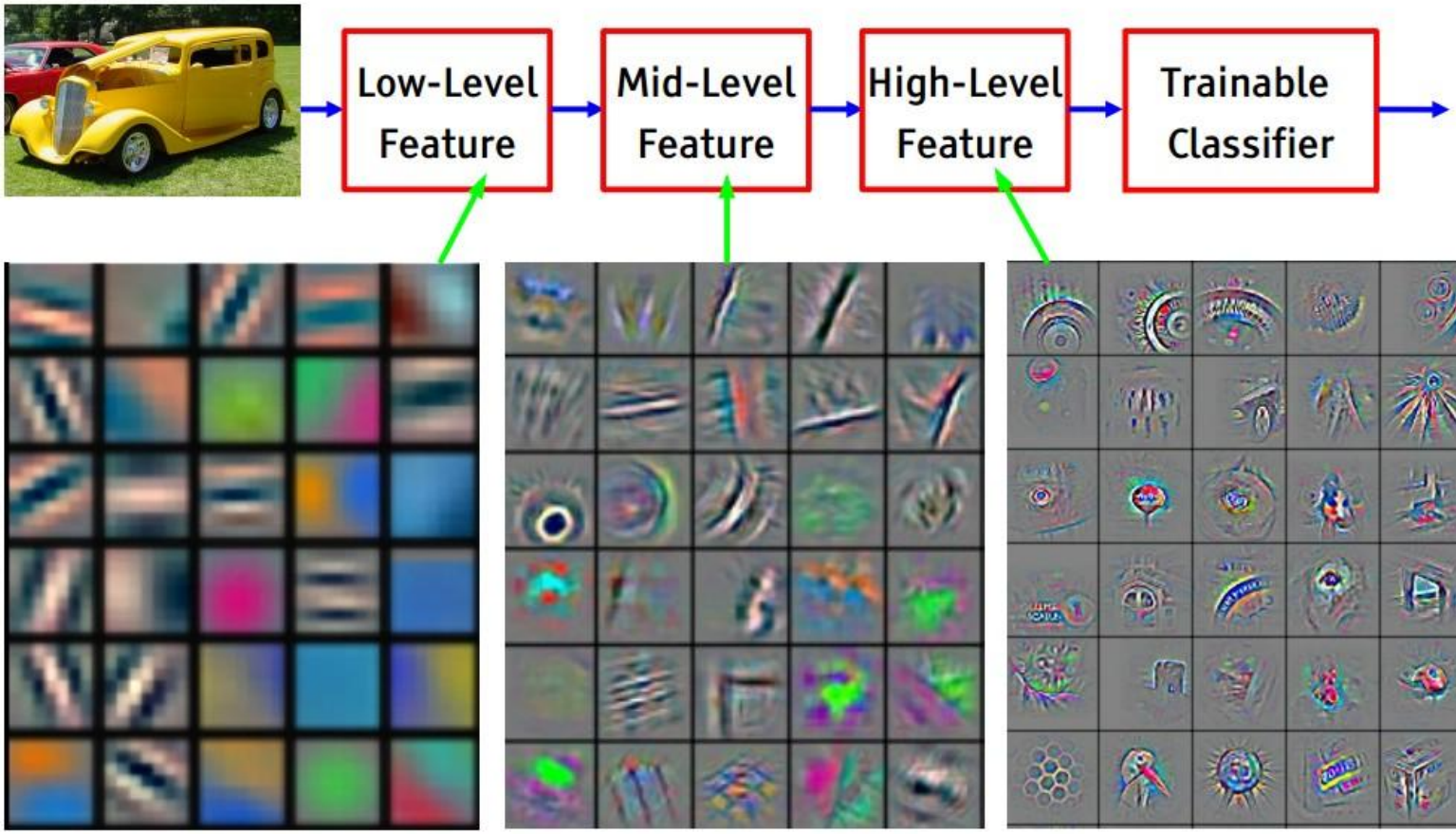


1x1 convolution layers

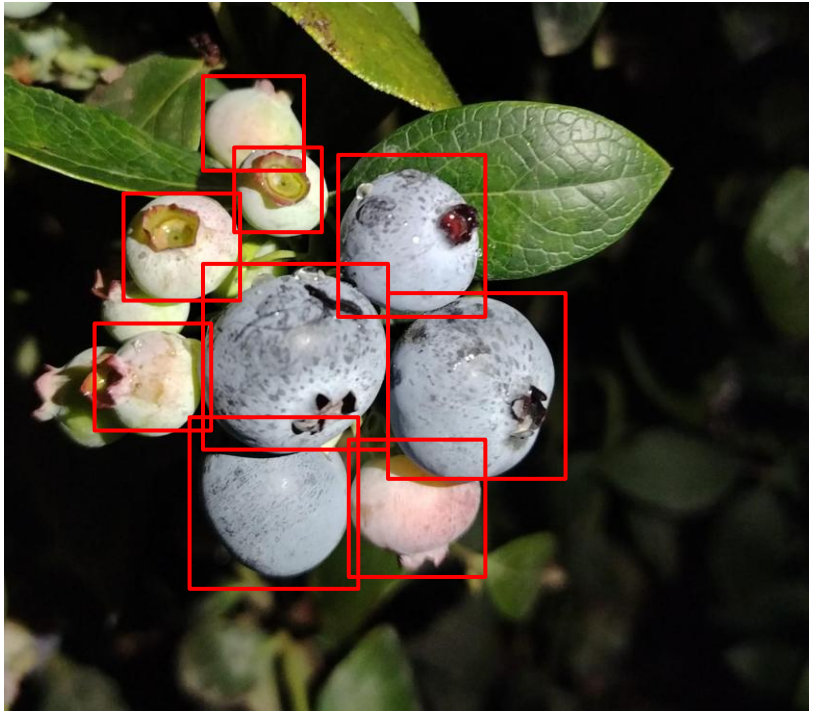
Ej: 128 filtros de 1x1 en un input de 1x1x64

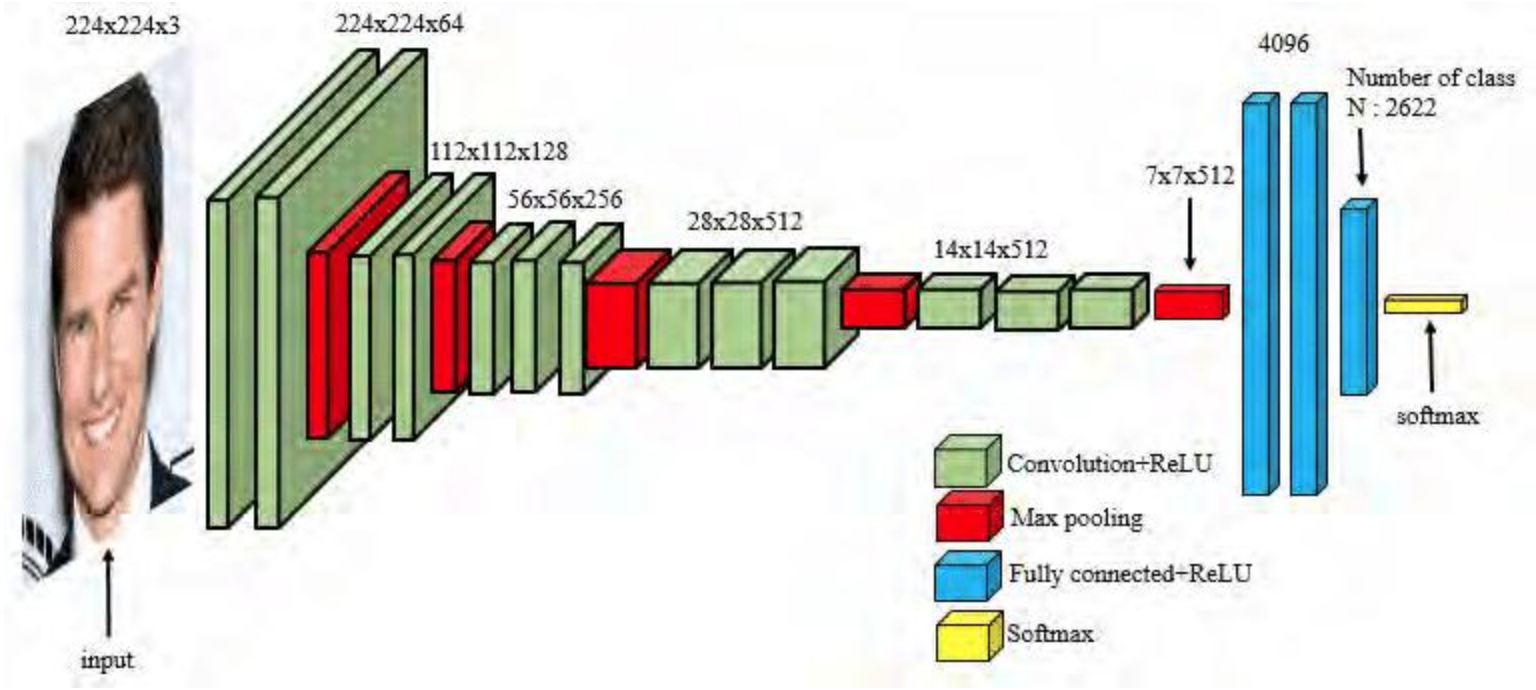


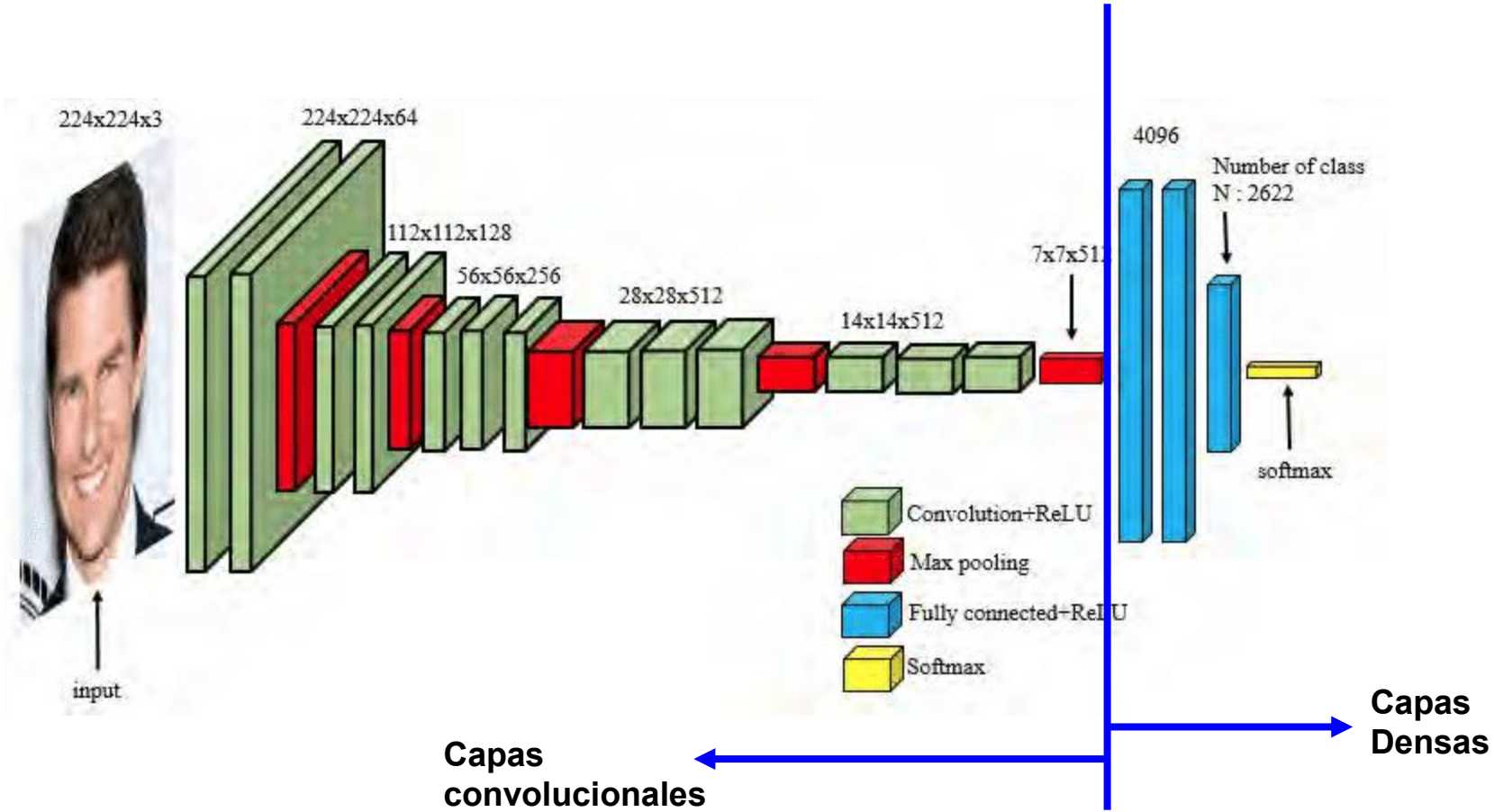
Transfer Learning

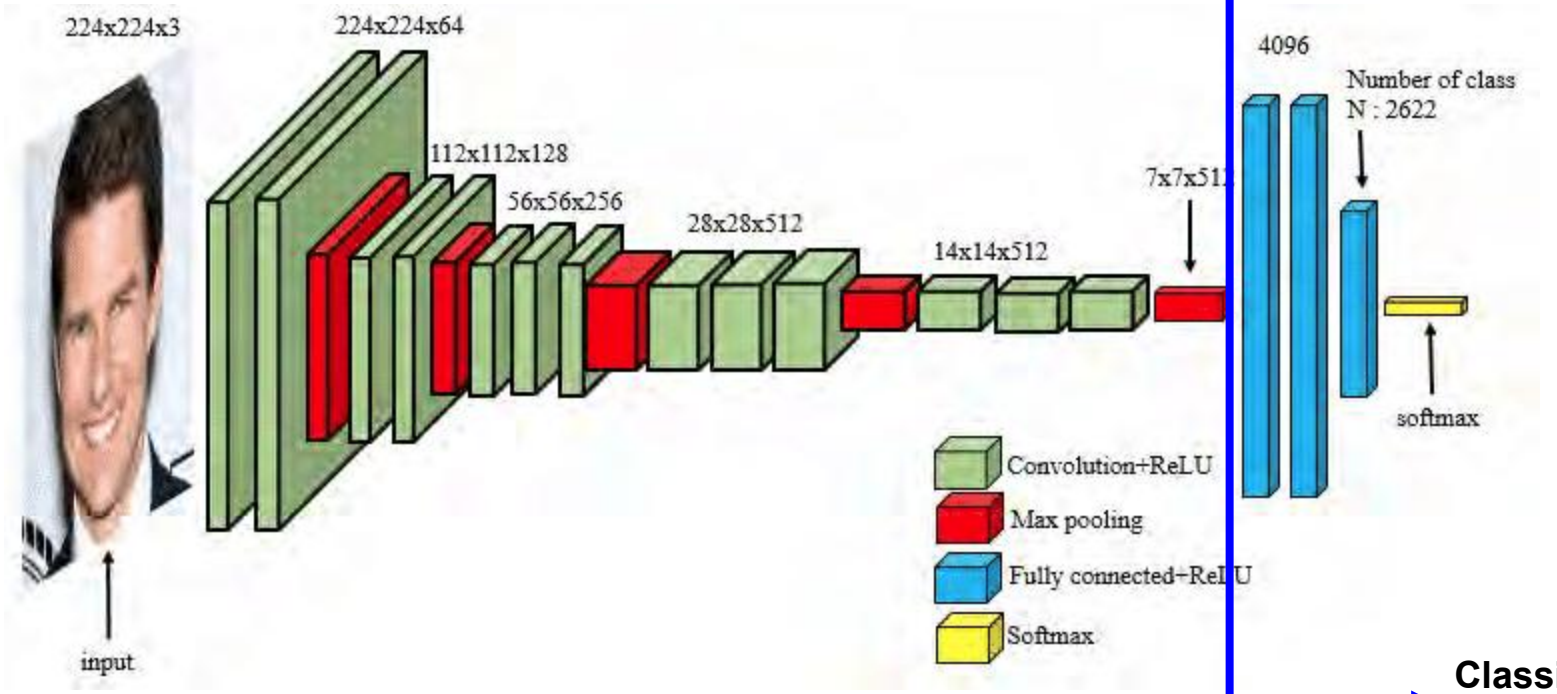


Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

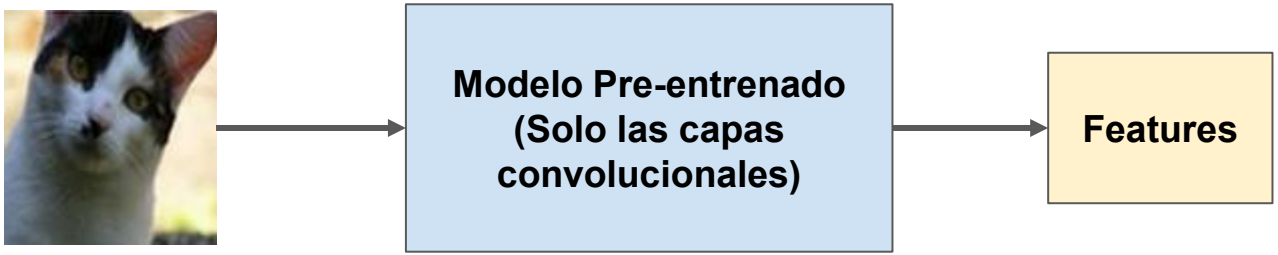



224x224x3
224x224x64
112x112x128
56x56x256
28x28x512
14x14x512
7x7x512
4096
Number of class
N : 2622
softmax
input
Convolution+ReLU
Max pooling
Fully connected+ReLU
Softmax

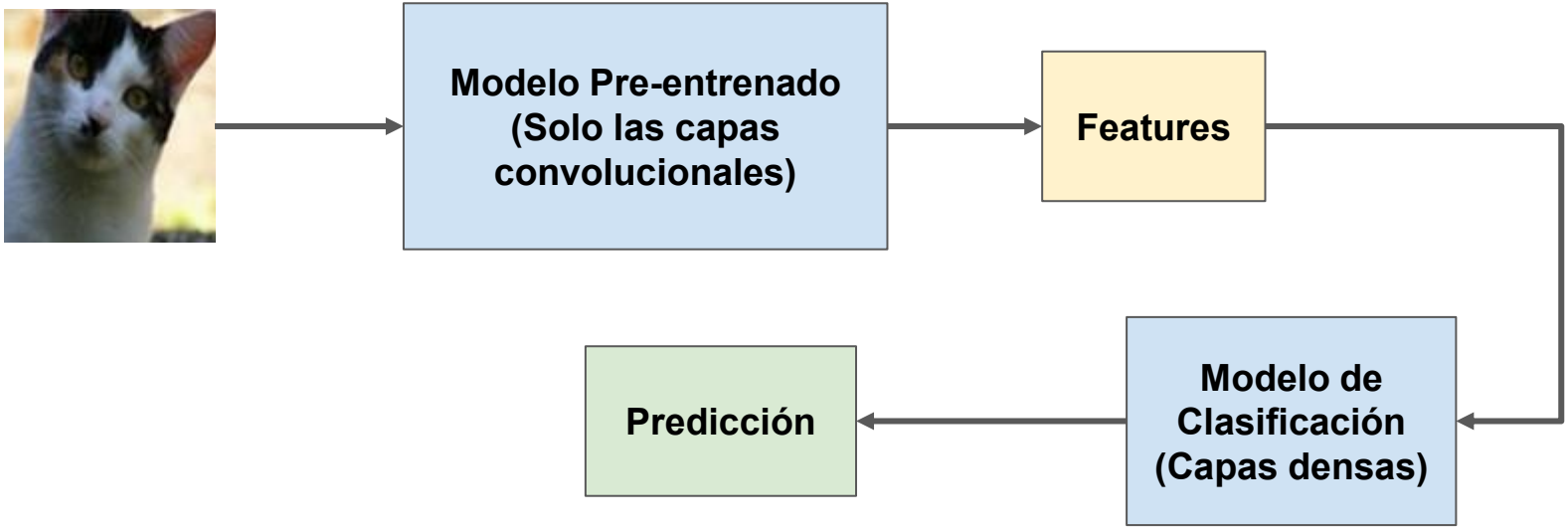

224x224x3
224x224x64
112x112x128
56x56x256
28x28x512
14x14x512
7x7x51
4096
Number of class
N : 2622
softmax
input
Convolution+ReLU
Max pooling
Fully connected+ReLU
Softmax
Capas convolucionales
Capas Densas


224x224x3
224x224x64
112x112x128
56x56x256
28x28x512
14x14x512
7x7x51
4096
Number of class
N : 2622
softmax
input
Convolution+ReLU
Max pooling
Fully connected+ReLU
Softmax
Feature extraction
Classification head

Transfer Learning



Transfer Learning



Transfer Learning - entrenamiento



Modelo Pre-entrenado (Solo las capas convolucionales)

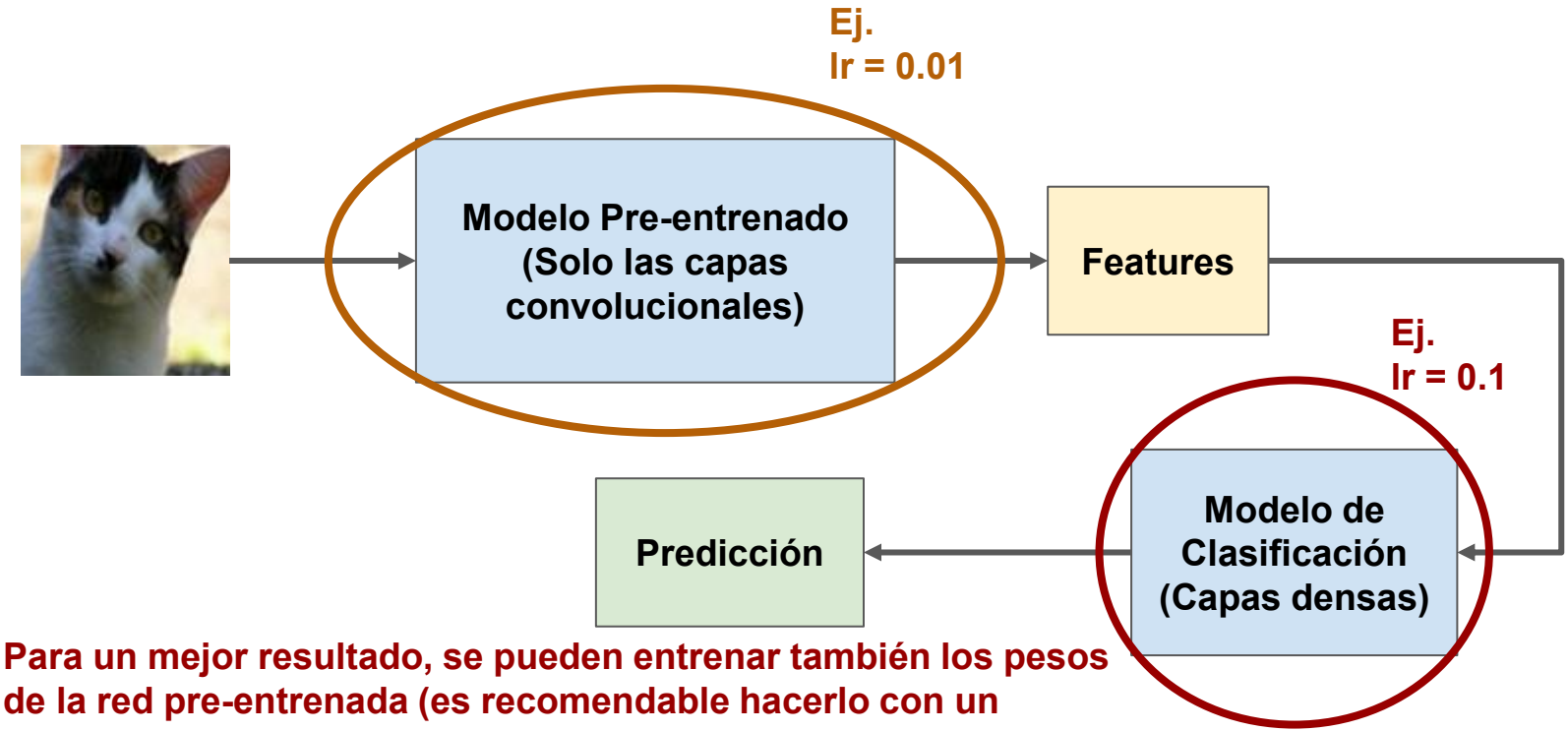
Features

Modelo de Clasificación (Capas densas)

Predicción

Se entrena el modelo de clasificación

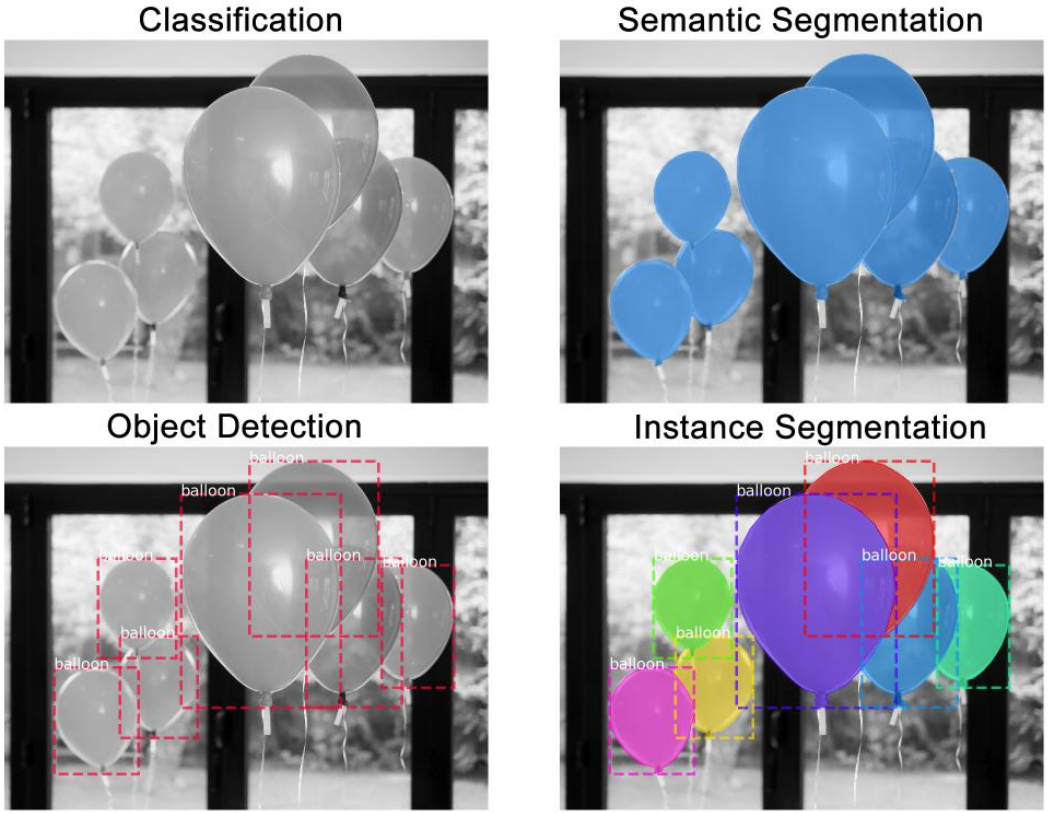
Transfer Learning - entrenamiento



Para un mejor resultado, se pueden entrenar también los pesos de la red pre-entrenada (es recomendable hacerlo con un learning rate menor: Learning-Rate-Multipliers-in-Keras).

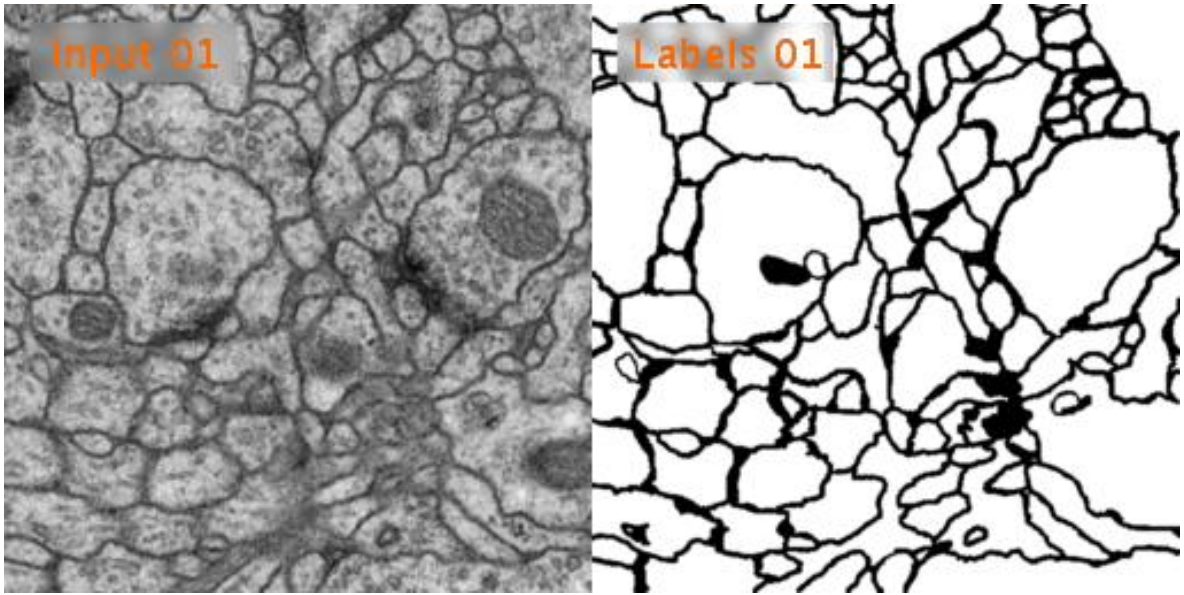
Aplicaciones

Detección de objetos



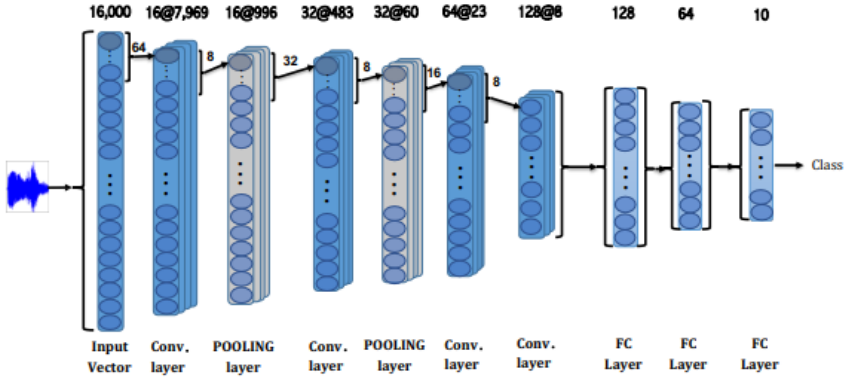
Fuente:https://engineering.matterport.com/splash-of-color-instance-segmentation-with-mask-r-cnn-and-tensorflow-7c761e238b46

Segmentación

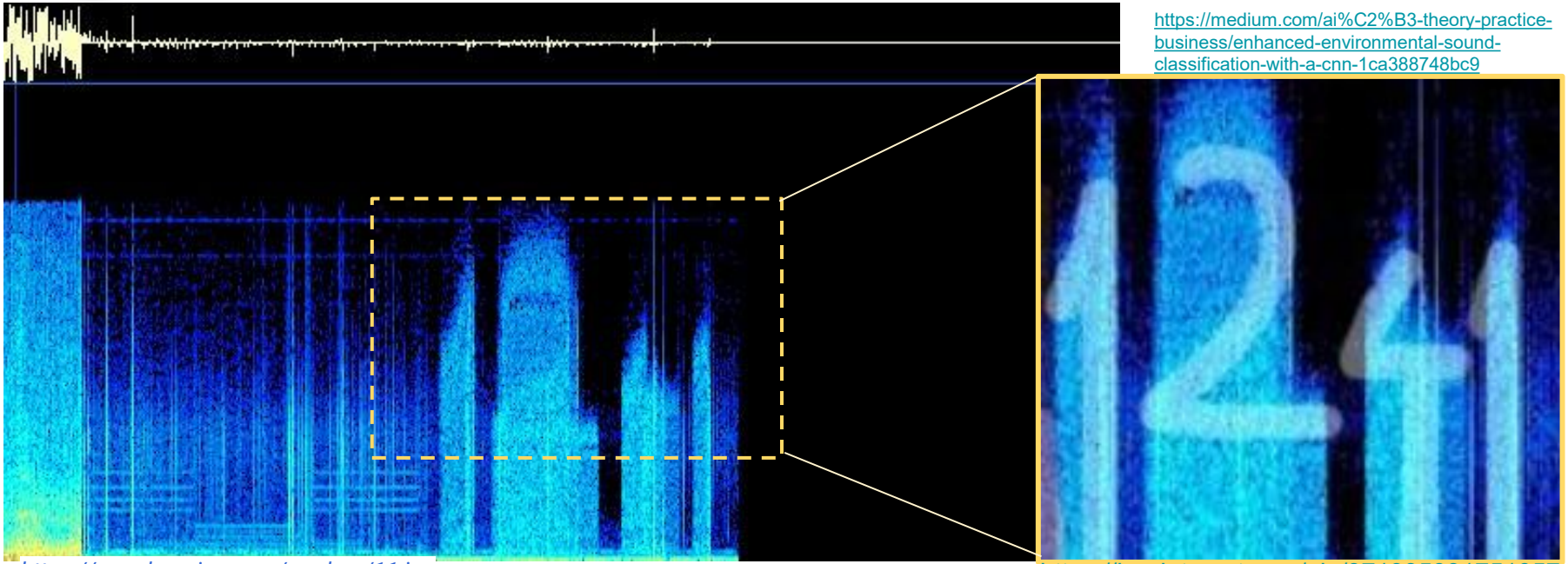


Fuente:https://towardsdatascience.com/review-u-net-biomedical-image-segmentation-d02bf06ca760

Clasificación de audio



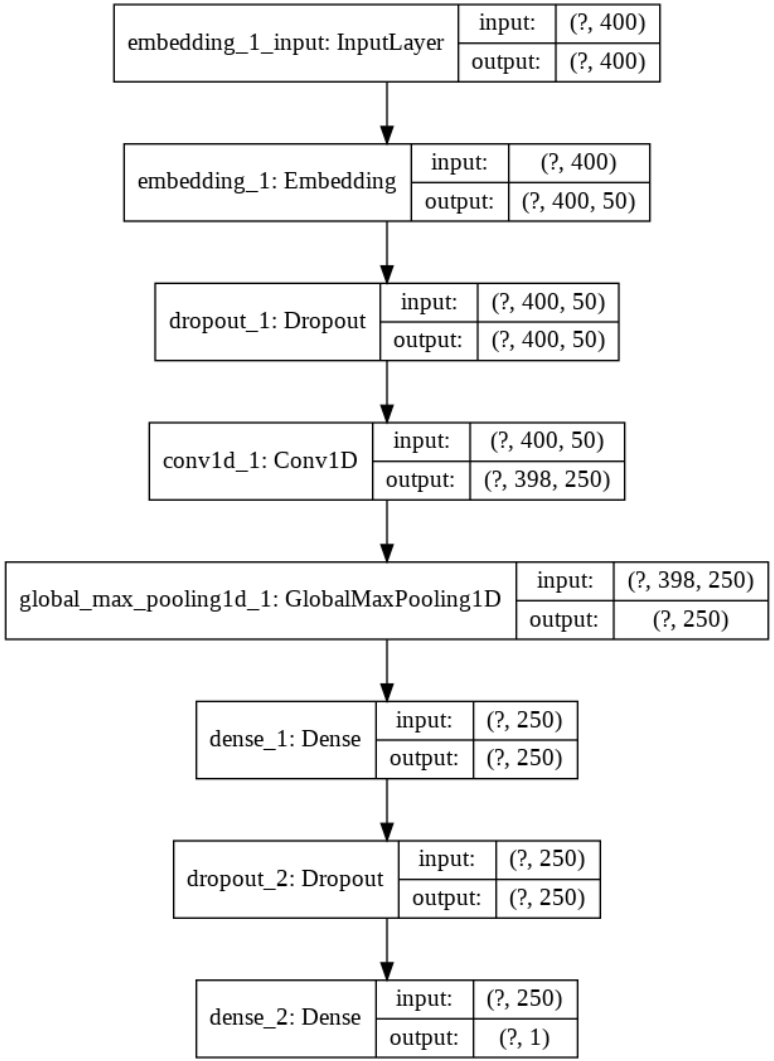
https://medium.com/ai%C2%B3-theory-practice-business/enhanced-environmental-sound-classification-with-a-cnn-1ca388748bc9



https://coredumping.com/random/11.jpg

https://br.pinterest.com/pin/371335931751057435/

Análisis de Sentimiento en texto



https://keras.io/examples/imdb_cnn/

Generación de imágenes



https://thispersondoesnotexist.com/





Imagined by a GAN (generative adversarial network) StyleGAN2 (Dec 2019) - Karras et al. and Nvidia
Don't panic. Learn how it works [1] [2] [3]
Help this AI continue to dream | Contact me
Code for training your own [original] [simple]
Art • Cats • Horses • Molecules | News | Friends | Office
Another | Save

Traducción de imágenes



https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix

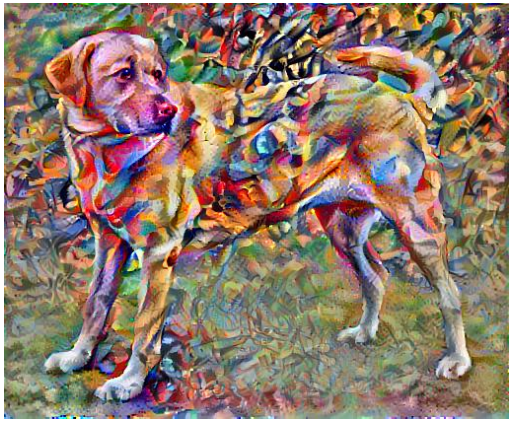
Transferencia de estilo



+



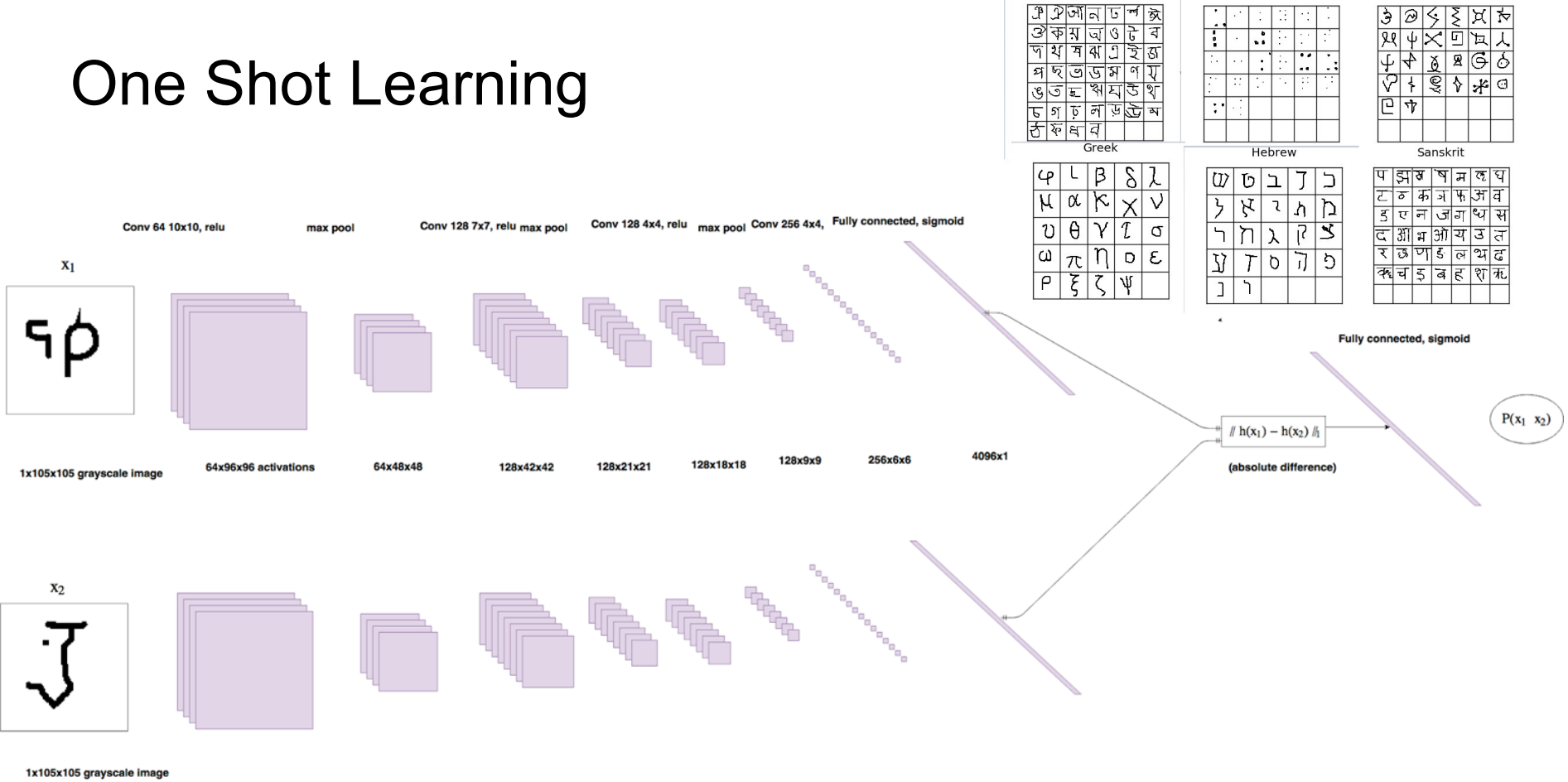
=



https://www.tensorflow.org/tutorials/generative/style_transfer

One Shot Learning

https://sorenbouma.github.io/blog/oneshot/



https://www.cs.cmu.edu/~rsalakhu/papers/oneshot1.pdf

Discusión


People with no idea about AI, telling me my AI will destroy the world
Me wondering why my neural network is classifying a cat as a dog..
Dog

Code time

https://www.kaggle.com/renatoh/keras-convnet-dl-pucp

https://www.kaggle.com/renatoh/keras-convnet-dl-pucp-weak-localization